

Project Planning Phase

Product Backlog, Sprint Schedule, Stories & Story Points – Online Complaint Registration

Project Name	Online Complaint Registration System (MERN Stack)	Date	4 August 2026
Maximum Marks	5 Marks		

Note: Date is a placeholder — update with your actual internship details.

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-1	Project Setup & Architecture	USN-1	As a team, we can design the ER diagram for Users, Complaints, and Departments.	2	High
Sprint-1		USN-2	As a developer, I can set up the project folder structure for client and server.	1	High
Sprint-1		USN-3	As a developer, I can initialize the Node/Express server with a basic route.	2	High
Sprint-1		USN-4	As a developer, I can configure the MongoDB connection using Mongoose.	2	High
Sprint-1		USN-5	As a developer, I can create Mongoose schemas for User, Complaint, and Department.	3	High
Sprint-1		USN-6	As a developer, I can set up the React app with routing and Bootstrap styling.	2	Medium
Sprint-1		USN-7	As a user, I can register for the application by entering email, password, and confirming password.	3	High
Sprint-1		USN-8	As a user, I can log into the application by entering email & password.	2	High
Sprint-1		USN-9	As a developer, I can implement JWT-based authentication middleware.	3	High
Sprint-2	Complaint Filing & Assignment	USN-10	As a citizen, I can file a complaint with a category and description.	3	High
Sprint-2		USN-11	As a citizen, I can view a complaint's detail page with its current status.	2	High
Sprint-2		USN-12	As an officer, I can view my assigned complaints queue.	3	High
Sprint-2		USN-13	As an admin, I can assign or reassign complaints to departments/officers.	3	High
Sprint-2		USN-14	As a citizen, I can filter/sort my complaints by status or date.	2	Medium
Sprint-2		USN-15	As a developer, I can implement role-based access control (citizen/officer/admin).	3	High
Sprint-2		USN-16	As a developer, I can encrypt passwords using bcrypt.	2	High
Sprint-2		USN-17	As a user, I can update my profile information.	2	Medium
Sprint-3	Tracking & Resolution	USN-18	As a citizen, I can attach photo/video evidence to my complaint.	3	High
Sprint-3		USN-19	As an officer, I can update a complaint's status (in progress/resolved).	3	High
Sprint-3		USN-20	As a citizen, I can cancel or reopen a complaint.	2	Medium

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint-3		USN-21	As an officer, I can add resolution notes/comments to a complaint.	3	High
Sprint-3		USN-22	As a citizen, I can view my complaint history and status timeline.	2	Medium
Sprint-3		USN-23	As a developer, I can implement real-time status sync across dashboards.	3	Medium
Sprint-3		USN-24	As a citizen, I can receive email/SMS confirmation of status updates.	2	Medium
Sprint-3		USN-25	As a developer, I can test complaint APIs using Postman.	2	Low
Sprint-4	Reporting, Admin & Deployment	USN-26	As an admin, I can view a department-wise complaint analytics dashboard.	3	High
Sprint-4		USN-27	As a citizen, I can submit a satisfaction rating after resolution.	2	Medium
Sprint-4		USN-28	As an admin, I can monitor officer performance and platform activity.	3	Medium
Sprint-4		USN-29	As a developer, I can optimize MongoDB queries for performance.	2	Low
Sprint-4		USN-30	As a developer, I can ensure a responsive UI across devices.	3	Medium
Sprint-4		USN-31	As a developer, I can deploy the application and prepare the demo video.	3	High
Sprint-4		USN-32	As a developer, I can conduct final UAT and fix reported bugs.	4	High

Project Tracker, Velocity & Burndown Chart (4 Marks)

Sprint	Total Story Points	Duration & Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days 24 Feb 2025	01 Mar 2025	20	01 Mar 2025
Sprint-2	20	6 Days 03 Mar 2025	08 Mar 2025		
Sprint-3	20	6 Days 10 Mar 2025	15 Mar 2025		
Sprint-4	20	6 Days 17 Mar 2025	22 Mar 2025		

Note: Sprint-1 is filled as a worked example. Update Story Points Completed and Actual Release Dates for Sprints 2–4 as your project progresses.

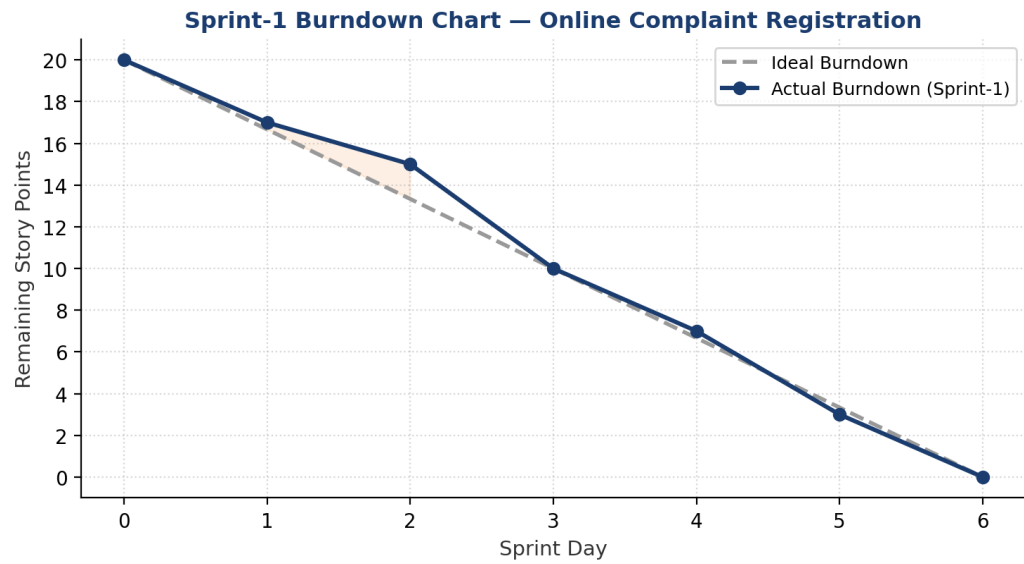
Velocity:

Each sprint runs for 6 days with a target velocity of 20 story points per sprint. The team's average velocity (AV) per iteration unit (story points per day) is calculated as:

$AV = \text{Sprint Duration} \div \text{Velocity} = 6 \div 20 \approx 0.3$ sprints per story point, or equivalently the team completes ≈ 3.3 story points per day on average.

Burndown Chart Description:

A burndown chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. Below is a burndown chart for Sprint-1 of the Online Complaint Registration project, comparing ideal vs actual progress.



References: visual-paradigm.com/scrum/scrum-burndown-chart | atlassian.com/agile/tutorials/burndown-charts | atlassian.com/agile/project-management | atlassian.com/agile/tutorials/epics | atlassian.com/agile/tutorials/sprints | atlassian.com/agile/project-management/estimation